Elections on measured stakes: plurality welfare tracking with microsimulated household impacts

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July 11, 2026

Abstract

Does plurality voting select the policy a stated social welfare function prefers when voters misperceive their own stakes? I study the question in a simulation whose inputs are measured and whose behavioral layers are explicit modeling choices: the electorate is 120,408 voting-age adults from certified, population-calibrated US microdata; the two platforms are actual encoded tax reforms with opposite incidence, budget-matched within 3%; each voter's stake is the tax engine's computed change in their household's net income, closed by a labeled financing rule; and perception — truth plus Gaussian noise plus optional shared bias — is the headline assumption. For plurality the model admits an analytic normal approximation at any electorate size, exact in the large-population limit. Measured stakes are two-cliffed: net of financing, three-quarters of adults hold under \$25 a year while over a fifth hold more than \$500, and this structure overturns standard toy-model intuitions. Tracking is non-monotone in accuracy: at perfect information the outcome is decided by the *sign* of a \$4.10-per-adult financing residual — a welfare-irrelevant quantity — while moderate noise ($\sigma \approx$ \$50–\$2,000) restores near-certain tracking by turning the trivial-stakes mass into canceling coin flips. Tracking thresholds are electorate-size statements that collapse toward 0.5 accuracy as n grows, while the expected-vote-share crossing behind the failure ceiling does not move with n. And shared misperception of $\approx$ \$221 per voter per year flips elections that \$1,000 of independent noise cannot move. The mechanisms are classical; the contribution is dollar-denominated calibration on measured microdata, a reusable open-source harness, and a direct path to replacing the perception assumption with survey estimates.

1 Introduction

A large literature asks when majority voting aggregates dispersed information or dispersed interests into good collective choices. The Condorcet jury theorem gives the optimistic benchmark: homogeneous stakes and independent errors let even barely-informed majorities decide almost perfectly (Condorcet, 1785). Its known failure modes are equally classical: correlated errors break the aggregation (Ladha, 1992), the crux of the miracle-of-aggregation debate (Page and Shapiro, 1992; Caplan, 2007); intense minorities lose to indifferent majorities under one-person-one-vote (Dahl, 1956); rationally ignorant voters may economize on information

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(Downs, 1957), and less-informed voters may strategically abstain in common-value settings (Feddersen and Pesendorfer, 1996) — a distinct mechanism from the private-stake indifference abstention modeled here; and probabilistic-voting models tie what elections aggregate to the responsiveness of vote probabilities (Coughlin and Nitzan, 1981; Lindbeck and Weibull, 1987).

What this paper adds is not a new mechanism but a new input quality. Simulation exercises in this tradition typically invent the key quantity — the distribution of what voters stand to gain or lose — and the invented distributions drive the conclusions. Here the inputs are measured and the behavioral layers are explicitly labeled. The electorate is population-calibrated microdata; the platforms are actual encoded reforms; the stakes are computed by a tax-benefit microsimulation engine, household by household, including interactions across programs. On these inputs, several toy-model conclusions reverse, and the surviving ones acquire dollar denominations: how many dollars of noise elections tolerate (thousands), how many dollars of shared bias they do not (about \$221 per voter per year), and how many dollars of stake the decisive voting bloc holds at perfect information (\$4.10 per adult per year, net of financing).

Five results follow. First, measured stakes are nothing like the homogeneous or Gaussian stakes of standard toys: 76% of adults have approximately \$0 at stake gross of financing, while a fifth hold stakes above \$500 (Section 4.1). Second, welfare tracking is non-monotone in perception accuracy, and its failure at perfect information is a knife-edge with a precise anatomy: the outcome becomes *welfare-independent*, set by the sign of the residual cost gap between the two policies (Section 4.2). Third, every tracking threshold is an electorate-size statement; a closed form places the Monte Carlo results on the full ladder from 101 voters to the large-population limit (Section 4.3). Fourth, elections in this environment are robust to large independent misperception and fragile to small shared misperception (Section 4.4). Fifth, moment-matched Gaussian comparators — even preserving impact–income covariance — reproduce none of this structure, so calibrating stake *shape* matters more than matching moments (Section 4.5).

The exercise is a thought experiment about one mechanism: self-interested voting on misperceived, engine-computed household impacts. It is not political science and predicts no elections; real voters weigh values, identity, and information far beyond their own net income (Bartels, 2005; Stantcheva, 2021). Platforms are labeled generically throughout.

2 Model

Electorate. Each voter i is a voting-age adult carrying their household's true impact δ_{ij} (dollars per year) under each policy $j \in \{A, B\}$, a survey weight, baseline household net income, and the household's adult count a_i. Voting is per adult; welfare counts each household's dollars once (contributions divide by a_i).

Financing. Engine impacts are gross: a tax cut shows only winners. The headline specification levies each policy's total cost equally per adult, making both policies budget-neutral by construction; a proportional-to-income levy and the unfinanced gross case are stress tests. Net stakes and the margin $m_i = \delta_{iA} - \delta_{iB}$ are defined after financing.

Welfare. An isoelastic social welfare function over household net income ($\eta = 1$, incomes floored at \$1,000) ranks the two policies; "tracked" means the election selects the *ballot-best* policy, the higher-ranked of the two. Under this metric both financed policies score below the status quo, which plurality does not offer; Section 4.2 returns to this.

Perception and voting. Voter i perceives $\hat{\delta}_{ij} = \delta_{ij} + b_j + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim N(0, \sigma^2)$ i.i.d. across voters and policies, and b_j a shared bias. Under plurality with abstention at exact perceived indifference, voter i votes for A with probability

$$p_i(\sigma, b) \;=\; \Phi\left(\frac{m_i - (b_B - b_A)}{\sigma\sqrt{2}}\right) \qquad (\sigma > 0), \tag{1}$$

while at $\sigma = 0$ the vote is deterministic: A if the shifted margin is positive, B if negative, abstention if exactly zero. The implementation also supports multiplicative attenuation of m_i, fixed to 1 throughout this paper. so an election of n voters sampled with probability proportional to weight reduces to a multinomial over (A, B, abstain) with population-weighted mean probabilities $(\bar{p}_A, \bar{p}_B, \bar{p}_0)$. With $g = \bar{p}_A - \bar{p}_B$, the probability that A wins is $\Phi\big(ng/\sqrt{n(\bar{p}_A + \bar{p}_B - g^2)}\big)$ by the normal approximation, degenerating to a step function of g as $n \to \infty$. All plurality results use this normal approximation (exact in the large-population limit) or Monte Carlo simulation that matches it; approval voting and instant-runoff variants are simulated directly, and with two policies instant-runoff coincides with plurality absent exact ties.

3 Data and calibration

The electorate comes from PolicyEngine's certified, population-calibrated Populace microdata for the United States (57,240 households; 120,408 adults representing 268 million), scored with `policyengine-us` 1.764.6 for policy year 2026. Policy A raises the federal Child Tax Credit base amount from \$2,200 to \$3,200; Policy B caps the top two marginal income tax rates at 34%. The engine scores them at \$39.1B and \$38.0B a year of household net income change (model totals including state-tax interactions, not budget scores) — within 2.9%, so neither platform is simply the larger transfer. Incidence is opposite: 9.9% of Policy A's dollars reach the top income decile versus 99.7% of Policy B's.

The committed data artifact carries full provenance (engine versions, certified dataset revision and hash, reform parameter dictionaries) and passes builder validations that reconcile every extracted quantity against the engine's own weighted aggregates (worst relative discrepancy 10^{-16}). Regression tests pin the artifact facts the results below depend on — most fragilely, the sign of the cost gap — so a future data rebuild cannot silently invert them.

4 Results

4.1 Measured stakes are two-cliffed

Gross of financing, 76% of adults see approximately \$0 from both policies; Policy A's gains arrive in per-child quanta of about \$1,000 for the 20% of households with eligible children. Financing hands the untouched majority a small signed stake: the two levies differ by \$4.10 per adult per year, so 75.5% of adults hold margins under \$25 while 19.9% (all in households with children) hold pro-A margins above \$500 and 2.6% hold pro-B margins below $-$\$500 (Figure 1). The weighted mean absolute margin, \$652, describes almost nobody.

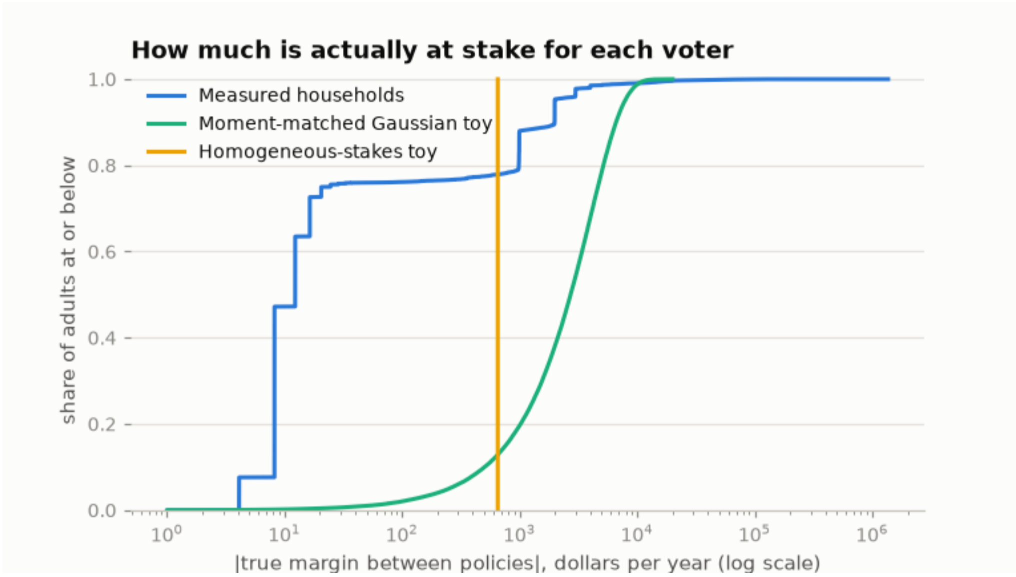

Figure 1: Weighted cumulative distribution of $|m_i|$, the true margin between the two policies per adult, net of per-capita financing. Measured households form a staircase — 47% of adults within \$10, 75.5% within \$25, driven by the \$4.10-per-adult difference in the two policies' levies — with a high-stakes tail. The two synthetic comparators used by toy models lack this structure.

4.2 Non-monotone tracking and the welfare-independence knife-edge

At 10,001 sampled voters, tracking is essentially perfect for noise σ between about \$50 and \$2,000 (every estimate 1.000, Wilson lower bounds 0.984) and collapses at both extremes (Figure 2). The high-noise decay is ordinary. The perfect-information failure is not: at $\sigma = 0$, 78.7% of adults strictly prefer Policy B — including the 76% of all adults untouched by either reform, whose entire stake is the \$4.10 levy gap — and plurality elects B in every election, the opposite of the welfare ranking.

Table 1 dissects the knife-edge. The direction is set entirely by the sign of the residual cost gap: rescaling Policy B to cost 3% *more* than A flips perfect-information tracking from 0% to 100% with the welfare ranking unchanged. The mass only rules while it votes: exact parity or unfinanced gross stakes leave it exactly indifferent, it abstains (turnout 24%), and tracking holds from accuracy ≈ 0.59 through perfect information. A behavioral \$25 indifference band does the same; a \$10 band does not, because multi-adult households' levy margins exceed it. Approval voting reaches the same outcome through self-silencing — voters who perceive both policies as net losses approve neither — though under the three-option ranking the welfare metric itself induces (status quo first), a winner elected on 23% turnout is still a welfare-reducing enactment; approval's property here is the filter, not welfare optimality.

Perfect information therefore does not make elections anti-track welfare; it makes them *welfare-independent*, delegating the outcome to whichever side of an arbitrary residual the no-stakes bloc shares. Between the extremes, moderate misperception helps: noise converts four-dollar margins into fair coins that cancel, and the informed high-stakes minority decides. This parallels the logic of probabilistic-voting models (Coughlin and Nitzan, 1981; Lindbeck and Weibull, 1987): idiosyncratic noise makes plurality weight stakes, and vanishing noise reverts it to counting heads — here traversed numerically on measured stakes.

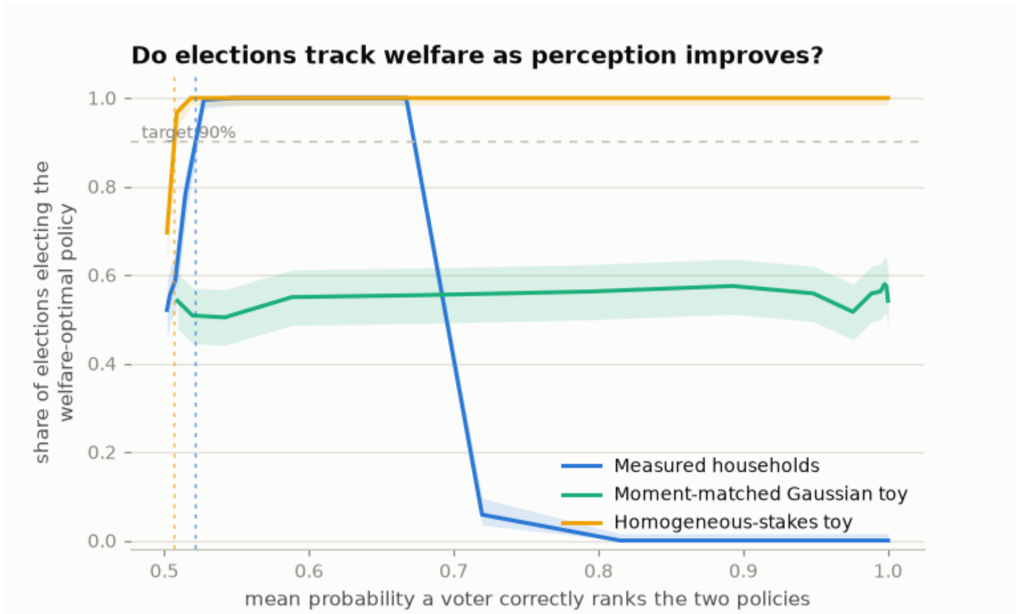

Figure 2: Share of 10,001-voter elections electing the ballot-best policy (240 elections per point; Wilson 95% bands), against mean ranking accuracy — the probability a voter correctly ranks the two policies for their own household. The homogeneous toy is the Condorcet jury theorem; the measured curve fails at *both* ends.

4.3 Thresholds are electorate-size statements

The accuracy band in which tracking exceeds 90% runs from 0.598–0.641 at $n = 101$ to 0.502–0.689 in the large-population limit (Figure 3). The entry threshold is Condorcet arithmetic and collapses toward 0.5 as n grows. The ceiling is pinned by an expected-vote-share crossing that does not move with n; the 90% boundary reaches it (≈ 0.69) from $n = 1{,}001$ onward, with only the smallest electorate (0.641 at $n = 101$) held below it by sampling noise. Any reported "accuracy threshold for elections to work" that does not state its electorate size is a sample-size artifact.

4.4 Bias, not noise, is the cheap attack

Holding $\sigma = \$1{,}000$ — where unbiased elections track essentially always — a shared misperception that Policy B is worth more than it is flips the median 10,001-voter election at $\approx \$221$ per voter per year (1.7% of elections flip at \$178, 97.5% at \$267). That is one-third of the mean absolute stake and one-fifth of a single per-child credit step. The contrast instantiates the correlated-errors breakdown of jury aggregation (Ladha, 1992) and the miracle-of-aggregation debate (Page and Shapiro, 1992; Caplan, 2007) in dollars: independent error in the thousands is harmless; shared error in the low hundreds is decisive.

4.5 Moment-matched toys reproduce none of it

A Gaussian electorate matched to the measured means and covariances of (impact$_A$, impact$_B$, log income) — preserving the incidence correlation that carries the welfare signal — has smooth,

Variant (perfect information)	Tracked	Turnout	Decisive margin
Per-capita financing (headline)	0%	100%	−\$4.10 levy gap
Proportional financing	0%	99.8%	same sign, income-scaled
Sign-flip counterfactual (B costs +3%)	100%	100%	+\$4.38 levy gap
Exact-parity counterfactual	100%	24%	\$0 (mass abstains)
Gross stakes (no financing)	100%	24%	\$0 (mass abstains)
Abstention band \$10	0%	53%	multi-adult margins survive
Abstention band \$25	100%	24%	staircase silenced
Approval voting	100%	23%	mass approves neither

Table 1: The perfect-information knife-edge under stress. Counterfactual rows are synthetic rescalings of Policy B's gross impacts, labeled as such in the regeneration scripts.

nearly centered margins: its perfect-information vote margin is a fraction of a percentage point. At 10,001 voters it never reaches 90% tracking; in the large-population limit it tracks with near-certainty. Neither behavior resembles the measured world's wide certainty band bounded by a knife-edge. The homogeneous-stakes toy (every voter holds the mean absolute margin) is the jury theorem, tracking from accuracy 0.507 at this n. What separates the measured world from both is not its first or second moments but its shape: the coexistence of a nearly indifferent majority with a concentrated high-stakes minority.

5 Robustness and limitations

The welfare ranking is stable across $\eta \in \{0.5, 1, 2\}$ and under per-capita and square-root equivalence scales, but flips to Policy B at $\eta \geq 3$, where the \$1,000 income floor's censoring of the poorest households' levies dominates; the ranking is a modeling choice and is reported as such. Both financing rules hand the no-stakes bloc its signed sliver; financing choices that zero it, or deficit financing that hides it, restore monotone tracking. Voters here are exclusively self-interested, static, and annual; candidates do not reposition (an endogenous-candidate extension is tracked in the repository). Errors are independent across voters — including adults in the same household — and across policies; household-correlated error is not simulated here and could change the levy bloc's effective behavior.

The binding limitation is that perception is assumed. The linear-Gaussian family is exactly what a survey regression of perceived on true impacts would estimate — attenuation, bias, and residual dispersion, by demographic cell — and the model's perception interface accepts a fitted empirical distribution without modification. Measured tax-policy knowledge, beliefs, and perceived distributional effects exist in the literature (Bartels, 2005; Stantcheva, 2021), but not the engine-computed, own-household incidence beliefs this model consumes; eliciting it is the natural next step, and its answer is a single point on the axes swept here.

6 Conclusion

On measured household stakes, the relationship between voter accuracy and welfare tracking inverts the toy intuition at both ends: elections track almost perfectly across two orders of magnitude of misperception, fail on a welfare-irrelevant residual at perfect information unless the trivial-stakes majority goes silent, scale their thresholds with electorate size, and trade a

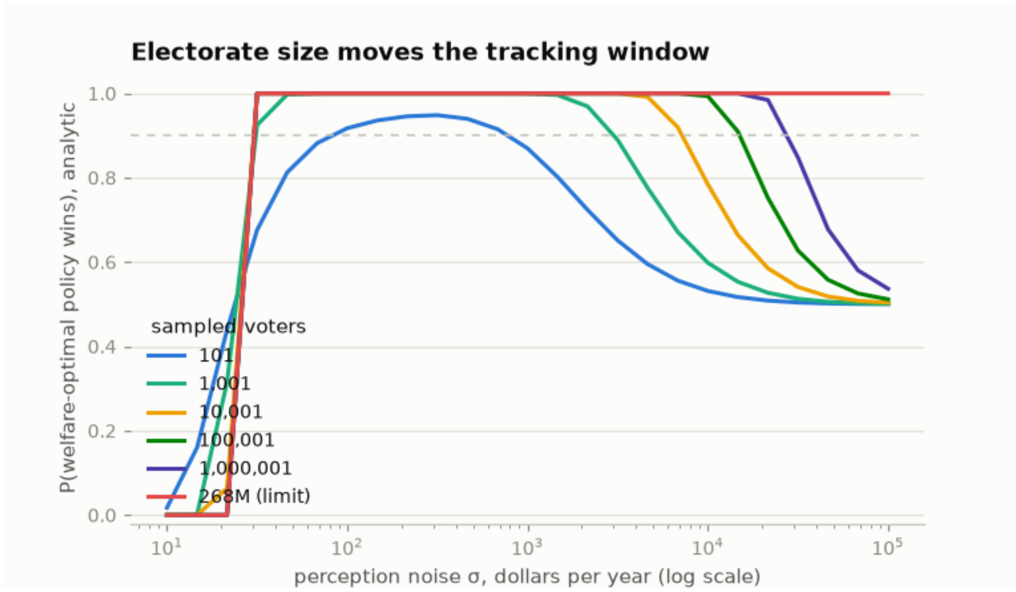

Figure 3: Analytic tracking probability by electorate size, per-capita financing, against noise σ (log dollars). On this axis the knife-edge is the fixed left wall near $\sigma \approx \$25$; the right slope, where noise finally drowns the high-stakes signal, is the jury-theorem edge that moves with n.

thousand dollars of noise tolerance for a two-hundred-dollar bias vulnerability. The mechanisms are old; the denominations are new. Whether real electorates sit on the tracking side of these numbers is now a measurement question.

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